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Efficient multiplex non-viral engineering and expansion of polyclonal $\gamma\delta$ CAR-T cells for immunotherapy

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Abstract

Gamma delta ($\gamma\delta$) T cells are defined by their unique ability to recognize a limited repertoire of non-peptide, non-MHC-associated antigens on transformed and pathogen-infected cells. In addition to their inability to mediate GvHD, $\gamma\delta$ T cells exhibit properties distinct from other lymphocyte subsets, prompting significant interest in their development as an off-the-shelf cellular immunotherapeutic. However, their low abundance in circulation, heterogeneity, limited methods for *ex vivo* expansion, and under-developed methodologies for genetic modification have hindered basic study and clinical application of $\gamma\delta$ T cells. Here, we implement a feeder-free, scalable approach for *ex vivo* manufacture of polyclonal, non-virally modified, gene edited chimeric antigen receptor (CAR)- $\gamma\delta$ T cells for therapeutic application. Engineered CAR- $\gamma\delta$ T cells demonstrate robust functionality *in vitro* and *in vivo*. Longitudinal *in vivo* pharmacokinetic

AUTHOR CONTRIBUTIONS

J.B., M.J.J., B.K., M.K., J.G.S., B.S.M., and B.R.W. conceived and designed the experiments. J.B., M.J.J., B.K., J.K., S.W., J.K., B.W., A.T.C., E.S., and J.G.S. conducted experiments and collected data. J.B., M.J.J., B.K., B.W., J.B., J.G.S., and B.R.W. performed data analysis. J.B., M.J.J., B.K., and B.R.W. wrote the manuscript.

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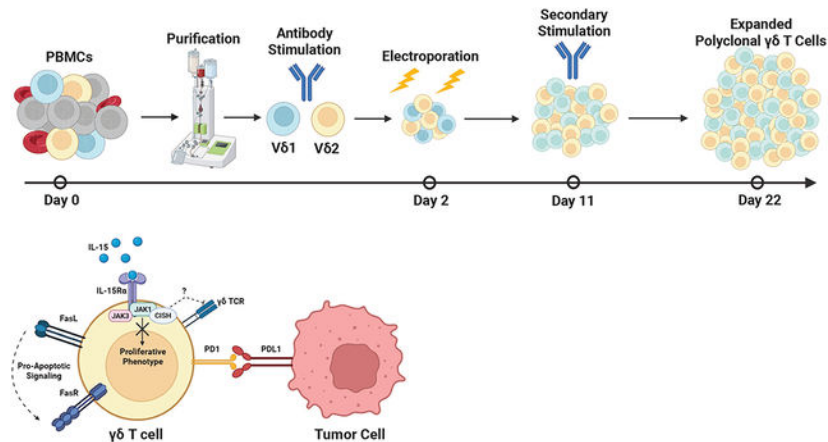
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DECLARATION OF INTERESTS

B.S.M. and B.R.W. have filed patents relating to the methods and approaches outlined in this manuscript and are founders and hold equity in Luminary Therapeutics who have licensed this IP.

profiling of adoptively transferred polyclonal CAR- $\gamma\delta$ T cells uncover subset-specific responses to IL-15 cytokine armoring and multiplex base editing. Our results present a robust platform for genetic modification of polyclonal CAR- $\gamma\delta$ T cells and present unique opportunities to further define synergy and the contribution of discrete, engineered CAR- $\gamma\delta$ T cell subsets to therapeutic efficacy *in vivo*.

Graphical Abstract



$\gamma\delta$ T cells' non-MHC antigen recognition and GvHD resistance make them ideal off-the-shelf immunotherapeutics. Bridge and colleagues developed a feeder-free, scalable protocol for non-viral multiplex base editing of polyclonal CAR- $\gamma\delta$ T cells, demonstrating robust *in vitro/in vivo* function. IL-15 armoring enhances subset specific persistence *in vivo*.

Keywords

$\gamma\delta$ T cells; Adoptive cell therapy; Chimeric antigen receptors; Base editing; Transposon integration; Polyclonal expansion; Cancer immunotherapy; Multiplex genome editing; Non-viral engineering; IL-15 armoring

INTRODUCTION

Gamma delta ($\gamma\delta$) T cells are a distinct subset of T cells characterized by their expression of heterodimeric pairs of gamma and delta T cell receptor (TCR) chains.¹ Unlike alpha beta ($\alpha\beta$) T cells, $\gamma\delta$ TCRs are not restricted to antigen presented on major histocompatibility complexes (MHCs) and therefore can directly bind unprocessed antigen including non-classical MHC molecules, members of the CD1 family, phosphoantigens, and stress-induced molecules, as well as small molecules and proteins associated with malignant transformation and pathogenic infection.^{2–7} Notably, several recent publications have demonstrated that phosphoantigens bind to butyrophilin proteins, which in turn, specifically activate V δ 2 $\gamma\delta$ T cells through binding the V δ 2V γ 9 TCR.^{8,9} $\gamma\delta$ T cells can also be activated via TCR-independent mechanisms including natural killer group 2D (NKG2D), a receptor shared by natural killer (NK), $\gamma\delta$, and $\alpha\beta$ CD8⁺ T cells that binds to MHC class I polypeptide-related sequence-A and -B (MICA, MICB) and UL16-binding proteins (ULBP1), all of which

display increased expression by transformed cells.^{10,11} Upon activation, $\gamma\delta$ T cells mediate their cytotoxic effects through the release of cytokines, chemokines, and cytolytic molecules, as well as upregulation of ligands to ‘death receptors’, such as Fas ligand and TRAIL.^{12–19}

The importance of $\gamma\delta$ T cells as a prognostic marker for cancer has been known for some time, with the presence of tumor-infiltrating $\gamma\delta$ T cells identified as a favorable clinical feature for patients with colorectal cancer, head and neck squamous cell carcinoma, and gastric cancer.^{20–22} Indeed, a meta-analysis of over 18,000 tumor samples from 39 different types of cancer identified $\gamma\delta$ T cell infiltration as the most favorable immune cell predictor of improved outcomes.²³ In the context of adoptive cell therapy (ACT), $\gamma\delta$ T cells offer several advantages over $\alpha\beta$ T cells or NK cells, including their ability to recognize a broad range of MHC-independent antigens, and the nearly absent risk of graft versus host disease (GvHD).^{24–27} The ability of $\gamma\delta$ T cells to function as professional antigen presenting cells (pAPCs) is often overlooked, particularly their ability to prime CD4+ and CD8+ $\alpha\beta$ T cells and amplify adaptive immune responses.²⁸

There are three prominent V δ alleles (i.e. *TRDV1*, *TRDV2*, and *TRDV3*), which code for V δ 1, V δ 2, and V δ 3, respectively.^{29,30} The V δ 2 subset predominates in peripheral blood and is the most studied subset, comprising 1–5% of all peripheral blood T cells and, of these 50–90% are paired with the V γ 9 allele (V γ 9V δ 2 T cells).^{31–33} V δ 1 $\gamma\delta$ T cells predominate within tissues including mucosa, tongue, intestine, lung, liver, and skin and have been shown to have superior migratory potential and function in hypoxic environments.^{34–38} V δ 1-V δ 2-CD3+ $\gamma\delta$ T cells, which include the V δ 3 subset, are a minor group that are found in blood, liver, and gut and have been shown to induce dendritic cell and B cell maturation.³⁹ Deniger et al, using feeder-expanded $\gamma\delta$ T cells, has shown that a population of polyclonal cells exhibits greater control than either V δ 1 or V δ 2 cell subsets alone in an ovarian cancer model.^{29,30}

Clinical application of $\gamma\delta$ T cells, including CAR- $\gamma\delta$ T cells, for ACT has lagged behind $\alpha\beta$ T cells and NK cells due to our limited understanding of their biology and a lack of effective methods for genetic modification and scalable *ex vivo* expansion. Zoledronate treatment, the most well studied method for expanding $\gamma\delta$ T cells, can produce large numbers of V γ 9V δ 2+ $\gamma\delta$ T cells that are effective *in vitro* against a variety of cancer cell lines, but does not expand other subsets.^{40–43} Several groups have used a combination of immunomagnetic separation or stimulation with anti-CD3 or $\gamma\delta$ TCR plate-bound antibodies to outgrow specific subsets or a combination of V δ 1 or V δ 2 subsets.^{44,45} Other approaches have used artificial antigen presenting cells to expand polyclonal $\gamma\delta$ T cells, but are limited by batch-to-batch inconsistency, a 40+ day production timeline, and regulatory hurdles including the need to establish master feeder cell banks.^{30,46} Additionally, current good manufacturing practice (cGMP)-compliant, feeder-free protocol using cytokines and anti-CD3 antibodies selectively outgrew V δ 1 T cells which demonstrated *in vitro* and *in vivo* anti-tumor cytotoxicity and robust production of IFN γ and TNF α .⁴⁶

$\gamma\delta$ T cells engineered to express a chimeric antigen receptor (CAR)^{47–49} have demonstrated 2–3 fold greater cell lysis vs unmodified $\gamma\delta$ T cells⁵⁰, improved on-target-on-tumor effect⁵¹, improved migration to tumors²⁸, and vigorous response to cancer cell lines *in vitro* and *in*

vivo, including tumor rechallenge models.⁵² Although the use of genome editing approaches like CRISPR/Cas9 or base editors to knockout (KO) checkpoint inhibitors (i.e. *PDCD1*, *CISH*) is a powerful approach to enhance T cell function^{53–55}, it is largely unexplored in $\gamma\delta$ T cells. CD8⁺ T cells edited to express a dominant negative FAS receptor (FAS^{DN}), evade Fas/Fas ligand signaling and demonstrate increased persistence *in vivo*.⁵⁶ Armoring $\gamma\delta$ CAR T cells to constitutively produce soluble IL-15 also increases persistence and survival *in vivo*.^{52,57}

Here, we report a scalable, non-viral, feeder-free approach for the manufacture of multiplex-base edited (BE) CAR- $\gamma\delta$ T cells, and demonstrate their application as an allogeneic cancer immunotherapy. This protocol involves two rounds of stimulation and ~22 days of expansion, achieving 15,000-fold expansion while maintaining TCR polyclonality. *TcBuster* transposon-based integration of α CD19-CAR and multiplexed BE demonstrates up to 50% CAR integration and 99% gene editing without selection or enrichment. Critically, this system can easily be adapted to incorporate a wide variety of larger, more complex constructs with alternative CAR or TCRs, cytokines/chemokines or their receptors^{58,59}, selection markers⁶⁰, or kill switches.^{61,62} Similarly, simply swapping gRNAs allows for targeting of alternative genes for KO or modification. Additionally, this approach allows uniform outgrowth of the V δ 1 and V δ 2 subsets instead of the subset-specific expansion achieved by existing methods. Thus, our highly flexible, current cGMP-compatible platform allows for high efficacy, non-viral integration of CAR and multiplex BE while retaining $\gamma\delta$ T cell subset diversity, reducing exhaustion, and increasing tumor-specific killing.

RESULTS

Antibody-Based Expansion of Polyclonal Peripheral Blood $\gamma\delta$ T Cells

We sought to develop a feeder-free method to expand $\gamma\delta$ T cells that was highly amenable to genome editing techniques. Given the well demonstrated ability of the α CD3 antibody clone OKT3 to strongly stimulate the expansion of human $\alpha\beta$ T cells and previous reports of antibodies specifically targeting the $\gamma\delta$ TCR, we wished to test their utility in protocols to non-virally engineer and edit $\gamma\delta$ T cells.^{44,45,63} We began by testing a panel of antibodies including α Pan- $\gamma\delta$ TCR, -V δ 1, -V δ 2, and a combination of -V δ 1, and -V δ 2 antibodies, as well as OKT3 and α CD16 antibody as controls in the presence or absence of α CD28 costimulatory antibody, to stimulate the expansion of $\gamma\delta$ T cells from PBMCs *in vitro* (Supplemental Figure 1A&B).

Stimulation of immunomagnetically sorted $\gamma\delta$ T cells with plate-bound α Pan- $\gamma\delta$ TCR antibody and soluble α CD28 antibody yielded 220-fold expansion, significantly greater than expansion with zoledronate, which produced an average of 90-fold expansion (Figure 1A). Additionally, unlike zoledronate, which relies on myeloid cells to produce reactive phosphoantigens, TCR agonist antibodies allow for subsequent restimulations without the need for accessory cells. Thus, we examined whether repeated antibody treatments could be used to enhance overall expansion. We found that restimulation on day 11 significantly enhanced outgrowth over a three-week timeframe, yielding a mean overall fold expansion of >14,000 by day 22 with little to no contamination with $\alpha\beta$ T cells (Figure 1B, Supplemental Figure 2A, and Supplemental Figure 3). Notably, cultures treated with α V δ 1 and α V δ 2

antibodies outgrew the V δ 1 and V δ 2 subsets respectively, whereas cultures treated with α Pan- $\gamma\delta$ TCR maintained the subset composition of the input population, with the same ratios of V δ 1, V δ 2, V δ 1-V δ 2-CD3+ $\gamma\delta$ T cell subsets (Supplemental Figure 2B&C).

Although the functional roles of distinct $\gamma\delta$ TCR subsets remain largely underinvestigated, current expansion protocols (zoledronate) promote outgrowth of only the V δ 2 subset.^{45,64} Thus, we examined whether antibody-based stimulation could maintain the diversity of V δ T cell subsets of peripheral blood $\gamma\delta$ T cells. Unlike zoledronate-treated cells, which were almost entirely V δ 2+ by day 11, populations stimulated with α Pan- $\gamma\delta$ TCR antibody maintained consistent V δ T cell subset frequencies across a 22 day expansion timeline, with no observable outgrowth of (Figure 1C and Supplemental Figure 4). TCR sequencing analysis confirmed that the majority of both γ - and δ -chain diversity was maintained across each 11 day and 22 day expansion phase (Figure 1D), demonstrating that antibody-based stimulation can support robust polyclonal expansion *in vitro*.

To analyze the impact of antibody-based TCR stimulation on $\gamma\delta$ T cell maturation, we stained for naive (CD45RO-CD27+), central memory (T_{CM}) (CD45RO+CD27+), effector memory (T_{EM}) (CD45RO+CD27-), and effector memory re-expressing CD45RA cells (T_{EMRA}) (CD45RO-CD27-) subsets.⁶⁵ The naive populations of V δ 1 and $\alpha\beta$ TCR-V δ 1-V δ 2- subsets decreased from 61.6% $\pm$ 2.6 and 55.1% $\pm$ 10.3 on Day 0 to 10.3% $\pm$ 11.3 and 6.3% $\pm$ 8.8 on Day 11 and 4.7% $\pm$ 4.3 and 2.6% $\pm$ 2.8 on Day 22, respectively (Figure 1E). Notably, our expansion protocol did not significantly impact either *TRD* or *TRG* sequence diversity or frequency (Supplemental Figure 5). Finally, we monitored the expression of many characteristic surface markers (Supplemental Figure 6A–O) over the course of our expansion protocol and found that the V δ 2 subset expressed lower levels of the coinhibitory molecules TIM3 and LAG3 than the either the V δ 1 or V δ 1-/V δ 2-/ $\alpha\beta$ TCR- subsets and that the V δ 1 subset expressed far less NKG2A than the V δ 2- and V δ 1-/V δ 2-/ $\alpha\beta$ TCR- subsets.

Taken together these data demonstrate that plate-bound α Pan- $\gamma\delta$ TCR antibody and soluble α CD28 antibody is an efficient means of rapidly expanding polyclonal $\gamma\delta$ T cells to therapeutically relevant levels, while maintaining subset and TCR clone diversity.

Efficient Transposon-Mediated CAR Integration in $\gamma\delta$ T Cells

After validating a scalable 3-week process for expanding polyclonal peripheral blood $\gamma\delta$ T cells, we sought to enhance and direct the innate anti-tumor activity of $\gamma\delta$ T cells by engineering them with an α CD19-CAR (Figure 2A, Supplemental Table 1). We previously developed and optimized a highly efficient non-viral CAR delivery approach in $\alpha\beta$ T cells and NK cells using a hyperactive *Tc Buster* (TcB, BioTechne) transposon system⁶⁶ and applied the platform to similar effect in $\gamma\delta$ T cells (Figure 2B). We found that the optimal electroporation time point for $\gamma\delta$ T cells occurs 48 hours post stimulation, resulting in a mean stable transposition efficiency of 47.3% $\pm$ 6.1 (Mean $\pm$ SD) vs electroporating at 72 hours post-stimulation (23.7% $\pm$ 5.4) or 96 hours post-stimulation (5.1% $\pm$ 1.3) (Figure 2C). Both the MTX selection (Figure 2D) and immunomagnetic enrichment using α CD19-CAR $\gamma\delta$ T cells transfected with a construct encoding RQR8⁶⁷ (Supplemental Figure 7A&B) approaches yielded >90% CAR+ $\gamma\delta$ T cells by day 22. Immunomagnetic enrichment did not impact production of the cytokines interferon gamma (IFN γ) or tumor necrosis factor (TNF)

following stimulation with PMA and Ionomycin (Supplemental Figure 7C). However, we do observe a slight reduction in IFN γ and TNF production following coculture with Raji target cells (Supplemental Figure 7D). Similarly, the frequency of CAR⁺ cells in non-enriched cultures did not change during expansion (Supplemental Figure 7E). We observed far superior fold expansion and integration using our pan- $\gamma\delta$ TCR-based stimulation protocol than with a zoledronate-based stimulation (Supplemental Figure 8A&B).

Overall expansion in transposon engineered $\gamma\delta$ T cells was 4,400-fold in the absence of selection and 2,600-fold in drug-selected populations (Figure 2E) without impacting day 22 viability or second round expansion (Supplemental Figure 9A–C). MTX selection also increased the average corrected vector copy number per genome from 1.99 ± 0.16 to 4.43 ± 0.29 (Figure 2F), both within FDA guidelines of 5 or less transgene copies per cell.⁶⁸ Finally, CAR integration occurred at similar efficiency across V δ 1, V δ 2, and $\alpha\beta$ TCR-V δ 1-V δ 2- subsets and the ratio of V δ subsets were not appreciably altered by construct delivery or selection method (Supplemental Figure 9D&E).

To evaluate whether α CD19-CAR $\gamma\delta$ T cells possess enhanced cytotoxicity, we performed a series of *in vitro* co-culture killing assays with target Raji-Luc cells. Because of the innate potential for non-engineered $\gamma\delta$ T cells to exhibit CAR-independent killing, we calculated E:T ratios based on total cell number, irrespective of the percentage of CAR⁺ cells. While control $\gamma\delta$ T cells largely failed to eliminate Raji-Luc cells, α CD19-CAR $\gamma\delta$ T cells completely cleared Raji-Luc cells after 24 hours at 3:1 and 1:1 E:T ratios and exhibited strong cytolysis even at 1:3 E:T ratios (Figure 2G). We also tested CAR $\gamma\delta$ T cells activity against a panel of tumor cell lines, including adherent cells and found significant activity against all (Supplemental Figure 10). Furthermore, Day 11 or Day 22 CAR- $\gamma\delta$ T cells exhibited similar cytotoxicity to Raji-Luc cells (Supplemental Figure 11). To further assess α CD19-CAR $\gamma\delta$ T cell function we performed serial killing assays to quantify sustained cytotoxic function and monitor exhaustion, in which α CD19-CAR $\gamma\delta$ T cells that successfully cleared Raji cells were re-plated with new target cells every 48 hours successively (Supplemental Figure 12). We found that CAR- $\gamma\delta$ T cells showed no loss of cytotoxicity throughout six rounds of killing (Figure 2H), indicating robust maintenance of cytolytic function. These data demonstrate that gene engineered CAR $\gamma\delta$ T cells can be efficiently manufactured and exhibit robust antitumor effector function *in vitro*.

Multiplex Base Editing of $\gamma\delta$ T Cells

In response to stimulation, $\gamma\delta$ T cells have been shown to upregulate FAS receptor at higher levels than $\alpha\beta$ T cells.⁶⁹ As a result, FAS-dependent cytotoxicity has been described as a mechanism of $\gamma\delta$ T cell activation induced cell death.⁷⁰ Provocatively, $\alpha\beta$ T cells engineered to express FAS^{DN} transgene have shown enhanced function and persistence after adoptive transfer in solid tumor models.⁵⁶ Although effective, dominant negative receptor engineering requires delivery of a cDNA.⁷¹ As an alternative, we devised an elegant strategy to utilize BE to disrupt FAS signaling by converting wildtype FAS to a FAS^{DN} (details of this will be reported separately).

To overcome activation-induced cell death in $\gamma\delta$ T cells and the immunosuppressive tumor microenvironment we tested multiplex knock out by combining our FAS^{DN} edit with

KO of *PCDC1*, a well characterized immune checkpoint, and *CISH*, a member of the suppressor of cytokine signaling (SOCS) family that decreases TCR functional avidity through degradation of PLC- γ 1 and provides negative feedback to gamma chain cytokine (ie IL-2, IL-4, IL-7, IL-9, IL-15, and IL-21) signaling (Figure 3A).^{53,72–75} Our α Pan- $\gamma\delta$ TCR antibody-based engineering protocol allowed for highly efficient BE (>90%) at all three loci (Figure 3B), even when performed contemporaneously with transposon-based CAR integration with minimal loss in cell viability (Supplemental Figure 13A–C).

In the absence of TCR stimulation, $\gamma\delta$ T cells engineered with FAS^{DN} exhibited a normalized survival of 80%, compared with 45% for CAR only controls (Figure 3C). The difference was even more striking following treatment with α Pan- $\gamma\delta$ TCR, as disruption of FAS receptor signaling yielded a six-fold increase in survival (Figure 3D).

To quantify the impact of PD1 KO on $\gamma\delta$ T cells, we exposed PD1 KO and control engineered α CD19-CAR engineered $\gamma\delta$ T cells to PDL1^{high} Raji-Luc cells (Supplemental Figure 14) in a killing assay. α CD19-CAR $\gamma\delta$ T cells containing a PD1 KO or a triple edit exhibited significantly greater killing than CAR-only controls after 24 hours (Figure 3E).

As previous reports found that CISH KO enhances $\alpha\beta$ T cell and NK cell responsiveness, we treated CISH KO $\gamma\delta$ T cells with decreasing concentrations of either IL-2 or IL-15 and assessed proliferation rates, which demonstrated that expanded CISH KO $\gamma\delta$ T cells achieved significantly greater outgrowth than untransfected pulse cells (Figure 3F).^{76,77} Loss of CISH trended towards an increase in IFN γ production in response to coculture with tumor cells, but this increase was not statistically significant. (Supplemental Figure 15). These observations parallel the effects of loss of CISH signaling in $\alpha\beta$ T cells and NK cells, where it acts as an important suppressor of cytokine signaling.^{73,74,78}

Finally, we combined all three base edits to assess their cumulative effect on α CD19-CAR $\gamma\delta$ T cell phenotype and function. Triple edited CAR- $\gamma\delta$ T cells performed equally as well as FAS^{DN}-only CAR- $\gamma\delta$ T cell in a FasL apoptosis assay (Figure 3C&D) and better than CAR- $\gamma\delta$ T containing only a PD1 KO in a killing assay at low E:T ratios (Figure 3E). Overall, these data demonstrate that targeted BE of negative regulators of $\gamma\delta$ T cells can increase their resistance to immunosuppressive signaling while enhancing $\gamma\delta$ T cell activation.

Engineered $\gamma\delta$ T Cells Display *In Vivo* Cytotoxicity

To investigate the potential of expanded polyclonal $\gamma\delta$ T cells for ACT, we utilized a well-established Raji-Luc murine xenograft model.^{55,79} Here, we tested groups with or without triple BE edits and with or without cytokine armoring with IL-15 expressing α CD19-CAR transposons (Figure 4A, Supplemental Figure 16A). We also included two extra groups that received two doses of the α CD19-CAR $\gamma\delta$ T cells seven days apart. Immunodeficient NSG (NOD.Cg-Prkdcscid Il2rgtm1Wjl/SzJ) mice were intravenously engrafted with 5×10^5 Raji-Luc cells four days prior to intravenous infusion using 10×10^6 expanded $\gamma\delta$ T cells manufactured in five technical replicates from a single human donor (Supplemental Figure 16B–F).

We observed tumor growth via bioluminescent imaging weekly for the duration of the experiment. Mice treated with $\gamma\delta$ T cells expressing the α CD19-CAR experienced suppressed tumor growth and lived twice as long as their control counterparts. (Figure 4B&C, Supplemental Figure 17A&B, and Supplemental Table 2).

To examine the distribution of engineered $\gamma\delta$ T cells within the treated mice, we performed necropsies as mice reached the predefined endpoint. We observed robust CD3⁺ lymphocyte infiltration, as measured by immunohistochemistry staining of human CD3 positivity in the skin, gut mucosa, and ovaries, as well as other tissues (Figure 4D, Supplemental Figure 18). Importantly, we also identified extensive CD3⁺ lymphocytes in the CNS, suggesting that these cells can cross the blood-brain barrier and may have potential as a therapeutic approach for tumors in immune privileged sites (Figure 4D).

Overall, our data demonstrate that $\gamma\delta$ T cells can be non-virally engineered to express a CAR with simultaneous multiplex BE without sacrificing performance or function following our cGMP-compatible α Pan- $\gamma\delta$ TCR antibody-based engineering process.

Dynamics of Engineered $\gamma\delta$ T Cell Subsets *In Vivo*

Following *in vitro* engineering and expansion, the ACT infusion products used for *in vivo* studies exhibited a typical peripheral blood $\gamma\delta$ T cell subset distribution characterized by >80% V δ 2 expression across all conditions (Supplemental Figure 7E). As our expansion protocol does not alter the frequency of $\gamma\delta$ T cell subsets, we monitored the dynamics of each $\gamma\delta$ T cell subset across the *in vivo* study via weekly blood draws and flow cytometry analysis. We examined the relative fraction of each major $\gamma\delta$ T cell subset within human CD3⁺ cells with the peripheral blood and the bone marrow (Figure 5A,C&E, Supplemental Figure 19), as well as their overall frequency within the peripheral blood and the bone marrow (Figure 5B,D&F).

We found a striking change in the relative frequency of distinct $\gamma\delta$ T cell subsets as a fraction of total CD3⁺ cells, changing from ~90% V δ 2⁺ and ~10% V δ 1⁺ at injection to ~50% V δ 2⁺ and ~40% V δ 1⁺ by week two, a trend that continued throughout the study (Figure 5A&B). This increase in V δ 1⁺ cells and decrease in V δ 2⁺ cells persisted at the endpoint in the majority of challenged mice, including those that received $\gamma\delta$ T cells without CAR engineering (Figure 5C, Supplemental Figure 20).

We also observed a major difference in the overall frequency of $\gamma\delta$ T cell subsets in peripheral blood between groups receiving $\gamma\delta$ T cells engineered with constructs encoding IL-15 increasing to over 10% of the overall PBMCs and those that did not express IL-15 remaining below 1% (Figure 5B&D). This increase occurred in all subsets with the V δ 1⁺ subset increasing ~80-fold and the V δ 2⁺ subset increasing ~30-fold between the first week post-infusion and endpoint (Supplemental Figure 21). However, despite this notable difference, there was no improvement in survival in groups that received IL-15 secreting cells versus those receiving CAR-engineered cells alone (Supplemental Figure 22).

Additionally we performed TCR sequencing on the infusion product and two of the mice belonging to the α CD19-CAR triple BE edit w/ IL15 cohort at endpoint. Notably, the

frequency of the top 10 TCR clones for both the *hTRG* and *hTRD* genes present in the infusion product are greatly reduced or absent from T cells in the blood and bone marrow (Supplemental Figure 23A&B, **center and right columns**). Similarly, the frequency of the top 10 genes present in the blood and bone marrow are also absent or greatly reduced in frequency when compared with the infusion product (Supplemental Figure 23A&B, **left column**). This coincides with an increase in overall $\gamma\delta$ T cell subset diversity in the blood and bone marrow after infusion (Supplemental Figure 24).

The inclusion of IL-15 had the largest impact on the V δ 1+ population, increasing both the fraction of V δ 1+ cells as a percent of CD3+ cells (Figure 5G) and as a percent of total PBMCs (Figure 5H). The decrease of V δ 2+ as a frequency of CD3+ in the presence of IL-15 did not reach significance, but the increase of V δ 2+ as a frequency of total PBMCs did.

Notably, the frequency of $\gamma\delta$ T cells expressing the α CD19-CAR, as measured by RQR8 expression, increased throughout the experiment in mice treated with cells engineered with an α CD19-CAR, cells engineered with an α CD19-CAR and a triple BE edit, and cells engineered with a CD19-CAR and IL-15 (Supplemental Figure 25), indicating that CAR+ cells may have an expansion or survival advantage in an *in vivo* tumor challenge model. However, mice treated with cells engineered with an α CD19-CAR, a triple BE edit, and IL-15 showed no significant change in frequency of cells expressing α CD19-CAR. This lack of change could be attributed to a lack of statistical power or ineffective synergy between the combination of genetic changes.

Taken together, these data demonstrate that CAR $\gamma\delta$ T cells generated with α Pan- $\gamma\delta$ TCR antibody-based engineering are functionally active *in vivo* and can be used to help control tumor growth. It also shows that IL-15 can greatly increase the baseline level of both V δ 1+ and V δ 2+ $\gamma\delta$ T cells in peripheral blood, with greater expansion of the V δ 1+ subset.

DISCUSSION

The role of $\gamma\delta$ T cells in cancer immunity has been the subject of intense research recently. Their ability to rapidly activate, produce proinflammatory cytokines, directly lyse target cells, and present antigen afford them a unique combination of anti-tumor capabilities. $\gamma\delta$ T cells can be activated directly through their TCRs, independent of MHC presentation or through receptors such as TLRs, Fc receptors, or other innate immune receptors. Their activating ligands can be peptides, but also include phosphoantigens and stress-related antigens such as heat shock proteins and MICA/MICB. Notably, the presence of $\gamma\delta$ T cells in the tumor is the only cell subset that correlates with reduced disease progression.^{21,23}

$\gamma\delta$ T cells possess many attractive features as a platform for ACT, including their lack of MHC restriction, potent cytotoxicity, and ability to act as pAPCs. Despite this, their development as a clinical option for the treatment of cancer has greatly lagged behind $\alpha\beta$ T cells and NK cells, in part a consequence of immature methods for isolation, engineering, and expansion. This lack of development is partially due to a lack of functional knowledge and familiarity, as $\gamma\delta$ T cells are far less well studied than $\alpha\beta$ T cells and NK cells, and

partially caused by practical issues, as $\gamma\delta$ T cells are a relatively rare subset in the peripheral blood and murine $\gamma\delta$ T cells are very different than human $\gamma\delta$ T cells.⁸⁰

One of their most compelling attributes, a unique functional diversity driven by distinct $\gamma\delta$ TCR combinations, is attenuated by existing engineering and expansion methods.^{27,45} Here we show an antibody-based, cGMP-compatible expansion protocol promotes the robust *ex vivo* outgrowth to clinically-relevant numbers of V δ 1, V δ 2, and V δ 1-/V δ 2-/ $\alpha\beta$ TCR-cells, and is amenable to CAR integration and multiplex genome editing using Cas9 base editors. Approaches that maintain polyclonal V δ subset diversity have been demonstrated to increase the tumor control following immunotherapy.³⁰ Notably, Deniger et al established a subset hierarchy, with polyclonal $\gamma\delta$ T cells being superior, followed by isolated V δ 1 cells, followed by V δ 1-/V δ 2-/ $\alpha\beta$ TCR- subsets, and with isolated V δ 2 cells being least able to control tumors.³⁰

This pan-expansion of $\gamma\delta$ T cells subsets also leads to the preservation of significant TCR diversity, enabling fundamental investigation into subset biology and inter-subset synergy. Furthermore, preservation of TCR repertoire during amplification allows for an opportunity to conduct scaled screenings for TCRs of desired specificity and pursue identification of novel cognate antigens and mechanisms of recognition in both normal and pathological states.

TcBuster transposon-based integration of CAR constructs yielded roughly 50% CAR+ cells with a copy-number rate of less than five copies per genome, with good antitumor potency even at low E:T ratios. Importantly, CAR $\gamma\delta$ T cells manufactured with this method did not show any reduction in potency over six rounds of serial killing *in vitro*, even at the lowest E:T ratios, suggesting that they are highly resistant to antigen-induced exhaustion. This is a major advantage in the setting of anti-tumor immunity as T cell exhaustion is a major contributor to loss of immune control of tumors.

Using BE, we successfully introduced precise edits at three genes (*CISH*, *PDCD1*, and *FAS*) simultaneously at rates of over 90%. PD1 KO increased the cells' ability to lyse PDL1-expressing tumor cells while CISH KO enhanced cell persistence in low cytokine environments. Whereas KO of PD1 and CISH was accomplished via splice site disruption, we instead chose to leverage the nucleotide editing capability of base editor to introduce a previously identified dominant negative mutation at the endogenous *FAS* locus, greatly reducing the cells' susceptibility to FAS/FAS ligand mediated suppression without the need to deliver an exogenous cDNA. We are currently adapting this approach to other suppressive surface proteins for which dominant negative mutations have been described, the results of which will be reported in greater detail separately.

Notably, the triple edit, no CAR group did not show improved survival or suppression of tumor growth compared with the PBS treated or pulse-control treated groups during a tumor challenge. Demonstrating the impact of these edits may require alternative solid tumor models, immunocompetent mouse models, or alternative gene targeted for KO or modification. However, our manufacturing protocol is readily adaptable to these additional experiments.

The adoptive transfer of α CD19-CAR $\gamma\delta$ T cells in mice seeded with CD19-expressing Raji cells slowed tumor growth and extended survival over mice treated with non-engineered $\gamma\delta$ T cells, indicating that CAR+ $\gamma\delta$ T cells can contribute to tumor control. CD3+ $\gamma\delta$ T cells were present in all tissues sampled, indicating their robust ability to traffic to and survey diverse tissues, including the CNS. Many CD3+ $\gamma\delta$ T cells were observed in the brain, demonstrating the ability of $\gamma\delta$ T cells to cross the blood-brain barrier and highlight their potential as a therapy for both primary tumors of the CNS and for other cancers that may metastasize to the brain. Importantly, no signs of GvHD or xenorejection were observed during the experiments. In general mice with higher tumor burdens experienced higher mortality, however we cannot completely rule out the possibility that some level of GvHD or other potential toxicity contributed to mortality.

The inclusion of soluble IL-15 “armoring” in CAR constructs had a significant impact on the outgrowth of $\gamma\delta$ T cells *in vivo*. $\gamma\delta$ T cells that were engineered with constructs expressing soluble IL-15 exhibited substantially increased frequency in the peripheral blood, with all treatment groups containing IL-15 having over 50% of all PBMCs being $\gamma\delta$ T cells. This increase was due to an expansion of both the V δ 1 and V δ 2 subset suggesting that IL-15 armoring may support cell engraftment and persistence in a clinical setting. However it is notable that the V δ 1 subset demonstrated greater outgrowth *in vivo* as measured by an increase in the frequency of V δ 1+ cells as a fraction of CD3+ cells and greater total fold-expansion in the peripheral blood. However, despite the major impact on cell outgrowth we observed no increase in survival in groups treated with $\gamma\delta$ T cells engineered with constructs containing IL15 over the groups treated with $\gamma\delta$ T cells engineered with constructs lacking IL15 (Supplemental Figure 22). The reasons for this lack of efficacy is unclear and it is possible that the additional cytokine and $\gamma\delta$ T cells do not provide any additional benefit above the baseline provided by CAR- $\gamma\delta$ T cells without IL15. Notably, in NK cell immunotherapies constitutive expression of IL15 paradoxically increases tumor control, but decreases overall survival, due to toxicity. While our therapy was generally well tolerated by mice in all groups, further examination for IL15 toxicity is necessary.

$\gamma\delta$ T cells are a group of versatile, multifunctional lymphocytes that hold great promise as an immunotherapy. Our cGMP-translatable manufacturing protocol supports further translation-focused efforts to evaluate their therapeutic potential. Further research is needed to determine the relative contribution and function of each $\gamma\delta$ T cell subset to an overall immune response, and whether specific genetic modifications are more well-suited to enhance the function of discrete subsets. Given $\gamma\delta$ T cells’ ability to interact and influence other immune cell types via proinflammatory cytokine production, ADCC, and antigen presentation on MHC Class I and Class II, future research using immunocompetent models is warranted to achieve the full potential of $\gamma\delta$ T cells as a robust cellular chassis for cancer immunotherapy and beyond.

MATERIALS AND METHODS

Donor $\gamma\delta$ T Cell Isolation and Expansion

Human peripheral blood mononuclear cells (PBMCs) were obtained from healthy, de-identified donors by automated leukapheresis (Memorial Blood Centers, Minneapolis,

MN, USA), then further purified via Ficoll-Hypaque (Lonza) separation. $\gamma\delta$ TCR⁺ cells were enriched by immunomagnetic negative selection using EasySep Human Gamma/Delta T Cell Isolation kit (STEMCELL Technologies). After purification, $\gamma\delta$ T cells were stimulated with either 5 μ M zoledronic acid (Sigma-Aldrich) or 10 μ g/mL plate-bound α Pan- $\gamma\delta$ TCR antibody (Miltenyi Biotec) and 2 μ g/mL soluble α CD28 antibody (ThermoFisher). Antibody-expanded populations were restimulated after 11 days using equivalent concentrations of each reagent. 10 μ g/mL of plate-bound α V δ 1 or α V δ 2 Abs (Miltenyi) were substituted for Pan- $\gamma\delta$ TCR when indicated to selectively outgrow specific subsets.

Cell Culture

$\gamma\delta$ T cells were maintained in OpTimizer T cell expansion media (Gibco) containing 10% human AB serum (Valley Biomedical), 2 mM L-glutamine (Gibco), and 100 U/mL Penicillin-Streptomycin (Millipore) and supplemented with 1,000 U/mL IL-2 (Peprotech), 5 ng/mL IL-7 (Peprotech), and 5 ng/mL IL-15 (Peprotech), referred to as “complete media” unless otherwise noted. Raji cells expressing luciferase (Raji-Luc cells), were thawed in RPMI 1640 (Cytiva) supplemented with 10% fetal bovine serum (R&D Systems) and 100 U/mL Penicillin-Streptomycin and then conditioned for growth in OpTimizer T cell expansion media (Gibco) containing 10% human AB serum (Valley Biomedical), 2 mM L-glutamine (Gibco), and 100 U/mL Penicillin-Streptomycin (Millipore). Cell lines were validated by the University of Arizona Genetics Core using STR profiling, and were routinely tested for mycoplasma contamination (Lonza). All cells were cryopreserved at 1×10^7 cells/mL in CryoStor CS10 (STEMCELL Technologies). All cell counts were performed using a Countess 3 (ThermoFisher), with trypan blue (Gibco) staining used to assess viability.

gRNA Design

Base editor guide RNAs targeting *CISH* and *PDCD1* splice donor sites were designed using the SpliceR webtool (<http://z.umn.edu/spliceR>). Six NGG gRNAs per target gene were evaluated in K562 cells, and the most efficient gRNAs (Supplemental Table 3) were purchased from Integrated DNA Technologies (IDT) with 2'-O-methyl and 3'-phosphorothioate modifications to the first three and last three nucleotides. A gRNA designed to introduce a *FAS* dominant negative mutation was also evaluated, mimicking the naturally occurring dominant negative FAS^{Y232C} mutation.^{81,82}

Electroporation of Activated $\gamma\delta$ T Cells for Transposon Integration and Multiplex BE

Unless otherwise specified, 48 hours after stimulation, $\gamma\delta$ T cells were pelleted and resuspended at a concentration of 4.5×10^7 cells/mL in T buffer (Neon Transfection Kit; Thermo Fisher Scientific) supplemented with an RNase inhibitor (Protector RNase Inhibitor, Roche 03335399001) at 267 U/mL. Hyperactive *Tc Buster* mRNA (1 μ g, TriLink Biotechnologies or Aldevron) and nanoplasid DNA encoding α CD19-CAR (1 μ g, Nature Technologies) were added to 3.6×10^5 cells on ice, followed by additional T buffer to a final volume of 12 μ L. Electroporations were performed with a 10 μ L Neon tip (Thermo Fisher Scientific) using three pulses of 1400 V and 10 ms width. To facilitate recovery, $\gamma\delta$ T cells were incubated at a concentration of 2×10^6 cells/mL in antibiotic-free complete

medium supplemented with DNase (1000 IU/mL) at 37 °C for 30 minutes. Following the recovery period, cells were resuspended to a final density of 1×10^6 cell/mL with complete media containing 200 U/mL Penicillin-Streptomycin. Control conditions included T buffer-only electroporations (“Pulse”) to account for electroporation effects and nanoplasmid-only conditions to monitor transient CAR expression. For multiplex gene knockouts, ABE8e Base Editor mRNA (1.5 µg, TriLink) and target gRNAs (1 µg) were added in addition to the nanoplasmid and Tc buster mRNA.

Selection of CAR+ Cells

For experiments using enriched CAR $\gamma\delta$ T cells, following nanoplasmid transfection stably-expressing CAR+ cells were enriched through one of two methods, depending on the selectable marker associated with each construct. CAR- $\gamma\delta$ T cells expressing an RQR8 motif were isolated by magnetic separation with α CD34 beads (Miltenyi), then assessed for RQR8 expression using flow cytometry. $\gamma\delta$ T cells engineered with mutant *DHFR*-containing CAR constructs were selected by adding 0.12 µg/mL methotrexate (MTX) to the culture media. Selected cells were exposed to MTX between days 5 and 11, unless otherwise specified, and were assessed for GFP expression by flow cytometry.

Antibodies and Flow Cytometry

Unconjugated α Pan- $\gamma\delta$ TCR (Clone: REA591; Miltenyi), α V δ 1 (Clone: REA173; Miltenyi), and α V δ 2 antibodies (Clone: B6; BioLegend) were used to expand $\gamma\delta$ T cells. CAR knock-in efficiency was quantified by GFP fluorescence and surface RQR8 expression. For flow eFluor 780-conjugated fixable viability dye (eBioscience) was used for live/dead discrimination. See Supplemental Table 4 for antibodies used in flow cytometry. Data were collected with a CytoFLEX S flow cytometer (Beckman Coulter) and all data were analyzed with FlowJo version 10.10.0 software (BD Biosciences).

Analysis of Gene-Editing Efficiency

Following engineering and expansion, genomic DNA was extracted using a GeneJet Purification Kit (ThermoFisher). To assess BE rates, PCR amplification was performed at the *CISH*, *PDCD1*, and *FAS* loci using primers (Supplemental Table 5) designed in Primer3Plus version 3.3.0⁸³. Following PCR purification with a QIAquick kit (QIAGEN), samples were outsourced for Sanger sequencing (Eurofins Genomics). The resulting .ab1 files were analyzed with EditR version 1.0.0 (<http://baseeditr.com/>).⁸⁴

CISH KO cytokine reduction Assays

Day 22-expanded *CISH*KO cells were thawed, resuspended in cytokine-free media, then supplemented with IL-2 or IL-15 at the indicated concentrations. Cultures were expanded for nine days, with cell count and viability recorded every three days.

Dominant-negative FAS Assays

FAS^{DN} and triple edited cells were thawed, resuspended in complete media, seeded at 1×10^6 cells/well in a 24-well plate, then rested overnight. Cells were then treated with recombinant Fas ligand (300 ng/mL; R&D Systems) and α His-tag antibody (10 µg/mL;

Clone: AD1.1.10; R&D Systems), and counted after 24 hours. A second set of KO samples was stimulated for 48 hours with α Pan- $\gamma\delta$ TCR and α CD28 prior to FasL treatment before use in *in vitro* killing assays.

Droplet Digital PCR (ddPCR)

ddPCR was used to assess the number of stably integrated CD19 CAR constructs following $\gamma\delta$ T cell transposition. Genomic DNA was extracted using a GeneJet Purification Kit (ThermoFisher). Primers and probes were designed using PrimerQuest online tool (Integrated DNA Technologies) using settings for 2 primers + probe quantitative PCR. Each sample was run as a duplexed assay consisting of an RNaseP internal reference primer + probe set (HEX) and a WPRE primer + probe set (FAM). Primers and probes were ordered from IDT (Supplemental Table 6). Reactions were set up using the ddPCR Supermix for Probes (no dUTP; Bio-Rad) with 200 ng genomic DNA per assay, according to manufacturer's instructions in technical triplicate. Droplets were generated and analyzed using the QX200 Droplet Digital PCR System (Bio-Rad). Thermocycler conditions were 95°C for 10 mins, followed by 40 cycles of 94°C for 30 seconds and 60°C for 1 min, ending with 98°C for 10 mins and 4°C hold. Frequency was visualized and quantified via the QuantaSoft version 14.0 software (Bio-Rad). Vector copy number of averaged triplicate was corrected through initial subtraction of background from non-engineered cells followed by normalization to the percentage of $\gamma\delta$ T cells expressing RQR8.

In Vitro Killing Assays

To assess the cytotoxicity of CAR- $\gamma\delta$ T cells, viably maintained Raji-Luc cells were seeded at 5×10^4 cells/well in cytokine-free $\gamma\delta$ T cell media across a black 96-well round-bottom plate. Cryopreserved $\gamma\delta$ T cells expanded out to day 11 or 22 were thawed, rested overnight, titrated in cytokine-free media, then cocultured at 3:1, 1:1, or 1:3 effector:target (E:T) ratios with Raji-Luc cells. Negative controls containing only Raji-Luc cells and positive controls treated with 1% Triton X-100 were also included. After normalizing media volumes to 200 μ L, cells were incubated for 24 hours, followed by addition of 20 μ L of 280 μ G/mL solutions of D-luciferin (10% in cytokine-free $\gamma\delta$ T cell media). Cytotoxicity was then assessed as a function of luminescence detected by a plate reader (BioRad).

Serial killing assays were also conducted at 3:1, 1:1, and 1:3 E:T ratios. All samples that displayed >90% cytotoxicity at the 3:1 E:T ratio were counted, titrated, then replated with new Raji-Luc cells every 48 hours for all E:T ratios. In all cases, samples at each E:T ratio were plated in technical triplicate.

In Vitro Stimulation and Functional Assays

Surface and intracellular cytokine staining (ICS) were performed following stimulation with various concentrations of plate-coated pan- $\gamma\delta$ TCR antibody and 2 μ g/mL of soluble CD28 antibody. To coat the plate with pan- $\gamma\delta$ TCR antibody, 100 μ L of 0, 0.5, 1, 2, 5, 10, or 20 μ g/mL of antibody solution was added in a 96-well plate, incubated overnight at 4 °C and washed twice with PBS before adding $\gamma\delta$ T cells that had previously been expanded for 22 days. $\gamma\delta$ T cells were seeded at 1×10^5 cells/well in RPMI containing 10% FBS and $1 \times$ Pen/Strep and supplemented with 2 μ g/mL of soluble CD28 antibody. For surface

staining, the cells were incubated for 24 hrs in a 37°C incubator and stained with appropriate antibodies. For ICS, the cells were incubated for 4 hrs with GolgiPlug (BD Biosciences) and monensin (eBioscience) with cells treated with an activation cocktail (BioLegend) as a positive control. After incubation, the cells were harvested, stained with eFluor 780 and appropriate surface markers to stain the $\gamma\delta$ T cells subset, and then permeabilized and fixed with cytofix/cytoperm solution (BD Biosciences) according to the manufacturer's instruction and BD Perm/Wash buffer (BD Biosciences) was used for all subsequent washes and incubations. To stain intracellular cytokines, eFluor 450-conjugated α INF- γ (Clone: 4S.B3; Invitrogen) and Brilliant Violet 510-conjugated α TNF- α (Clone: MAB11; BioLegend) were added and incubated for overnight at 4°C. Data were acquired on a CytoFLEX S flow cytometer and analyzed using FlowJo software version 10.10.0 (BD Biosciences).

***In Vivo* Tumor Challenge**

NSG mice were injected intravenously (IV) with 5×10^5 Raji-Luc cells. Four days later, control and engineered $\gamma\delta$ T cells were thawed, normalized to 50% CAR positivity by immunomagnetic selection for RQR8 (StemCell) and addition of pulse or pulse BE-edited cells, and then injected via tail vein at 10×10^6 cells/mouse. Select groups received a second infusion of engineered $\gamma\delta$ T cells seven days after the first infusion as indicated. Peripheral blood was collected weekly through the submandibular vein and tumor size was assessed via IVIS tumor imaging (Revvity). Peripheral blood, bone marrow, and other tissues (see histology) were collected at the endpoint from the longest surviving mice. Blood samples were analyzed by flow cytometry and an IL-15 ELISA kit (Abcam) was used to measure peripheral blood IL-15 concentration. All tumor growth was monitored by weekly bioluminescence imaging of mice 5 minutes post intraperitoneal (IP) injection of D-luciferin (100 μ L total volume, 28 mg/kg) using an IVIS100 imager followed by region of interest (ROI) analysis of images (Living Image software, version 4.7.3).

Immunohistochemistry (IHC)

All tissues were fixed with 10% formalin, paraffin embedded, sectioned, and stained with anti-CD3 (Clone: SP7, Novus Biologicals). CD3 stain was detected using 3,3'-Diaminobenzidine (DAB) in the presence of HRP and counterstained with hematoxylin. Slides were imaged using Proscia software, following previously described methods with slight modifications.⁸⁵

TCR Sequencing

Total RNA was purified from $0.5\text{--}1 \times 10^6$ $\gamma\delta$ T cells using the RNeasy Plus mini kit (Qiagen) according to the manufacturer's instructions and the TCR library was prepared using an iRepertoire arm-PCR platform- RNA-iR-Complete Dual Index Primer Kit (iRepertoire). Briefly, reverse transcription of 100 ng of total RNA was performed using the iR-RT-PCR1 enzyme mix. Then the PCR product was purified using selection beads followed by secondary amplification of the product using iR-PCR2 allowing for addition of Illumina adapter sequences. Finally, libraries were purified with selection beads and sequenced using Illumina MiSeq using 2X150 paired-end read length (MS-102–2002, Illumina). The TCR sequence FASTQ files were uploaded to the iRepertoire bioinformatics

website for analysis. Shannon entropy, a measure of TCR diversity, was calculated as described previously.⁸⁶

Statistical Analysis

Student's t-tests were performed to identify significant differences between two groups and one- or two-way ANOVAs followed by Tukey's post hoc test were used to evaluate differences between three or more groups unless otherwise indicated. The significance level was set at $\alpha = 0.05$. Mean values $\pm$ standard deviation are shown. Individual data points represent independent human donors unless otherwise noted. Statistical analyses were performed in GraphPad Prism version 9.5.

Supplementary Material

Refer to Web version on PubMed Central for supplementary material.

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DATA AVAILABILITY STATEMENT

TCRseq sequencing data that support the findings of this study will be deposited in the Gene Expression Omnibus (GEO) repository with the GEO accession number listed in this section prior to publication. Source data are available from the corresponding authors upon request.

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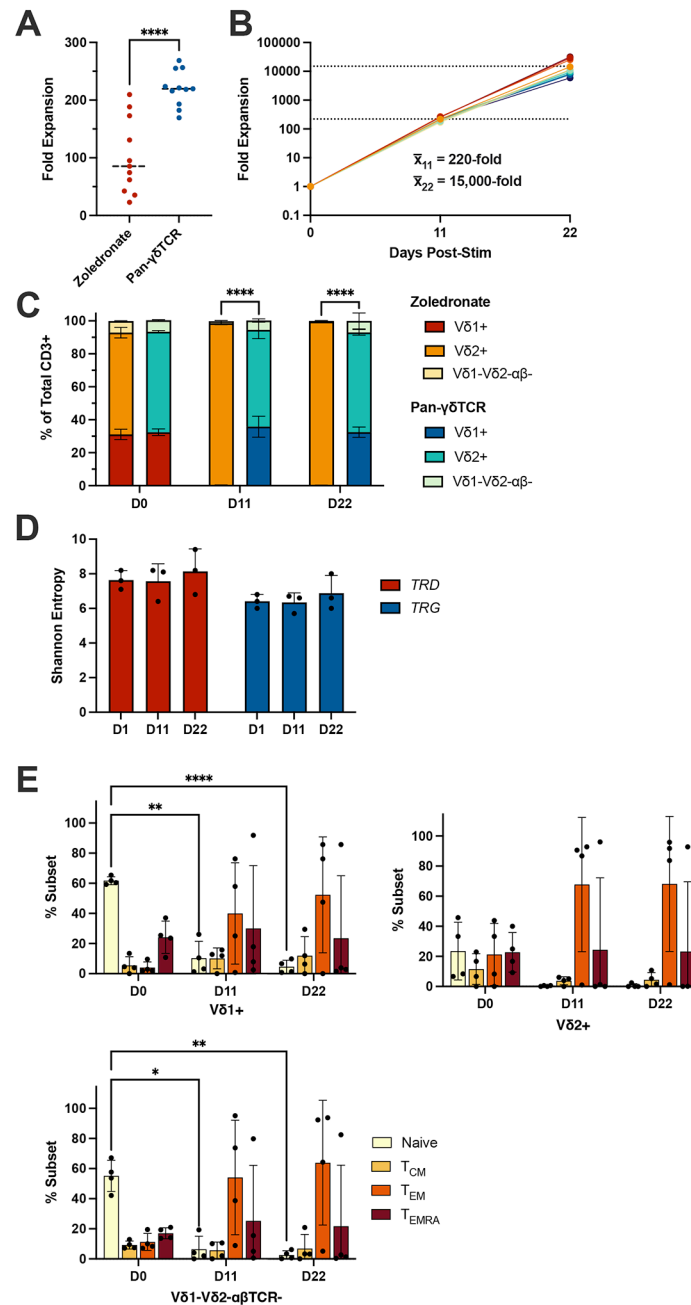


Figure 1. Antibody-based stimulation yields robust outgrowth of polyclonal $\gamma\delta$ T cell populations.

(A) Fold expansion after 11 days. Human $\gamma\delta$ T cells isolated from the peripheral blood were stimulated with either zoledronate or plate-bound α Pan- $\gamma\delta$ TCR and soluble α CD28 (n=11, paired T test). (B) Fold expansion at day 22 after two rounds of stimulation with α Pan- $\gamma\delta$ TCR and soluble α CD28 antibodies, occurring on day 0 and day 11 (n=11). (C) $\gamma\delta$ T cell subset frequency, as measured by flow cytometry, in zoledronate or α Pan- $\gamma\delta$ TCR plus soluble α CD28 stimulated $\gamma\delta$ T cells and days 0, 11, and 22 (n=4 independent biological donors, Two-way ANOVA with Tukey's multiple comparison test). (D) Bar graphs showing Shannon entropy, a measure of TCR clonal diversity, of $\gamma\delta$ T cell TCR diversity for TRD

chains (red bars) and TRG chains (blue bars) at day 0, 11, and 22 (n=3, Two-way ANOVA with Šidák's multiple comparison test). (E) Naive, T_{CM}, T_{EM}, and T_{EMRA} frequencies in Vδ1+ (*top left panel*), Vδ2+ (*top right panel*), and Vδ1-Vδ2-αβ- (*bottom panel*) γδ T cells subsets as determined by CD27 and CD45RO expression (n=4, Two-way ANOVA with Tukey's multiple comparison test). (****p<0.0001, **p<0.01, and *p<0.05)

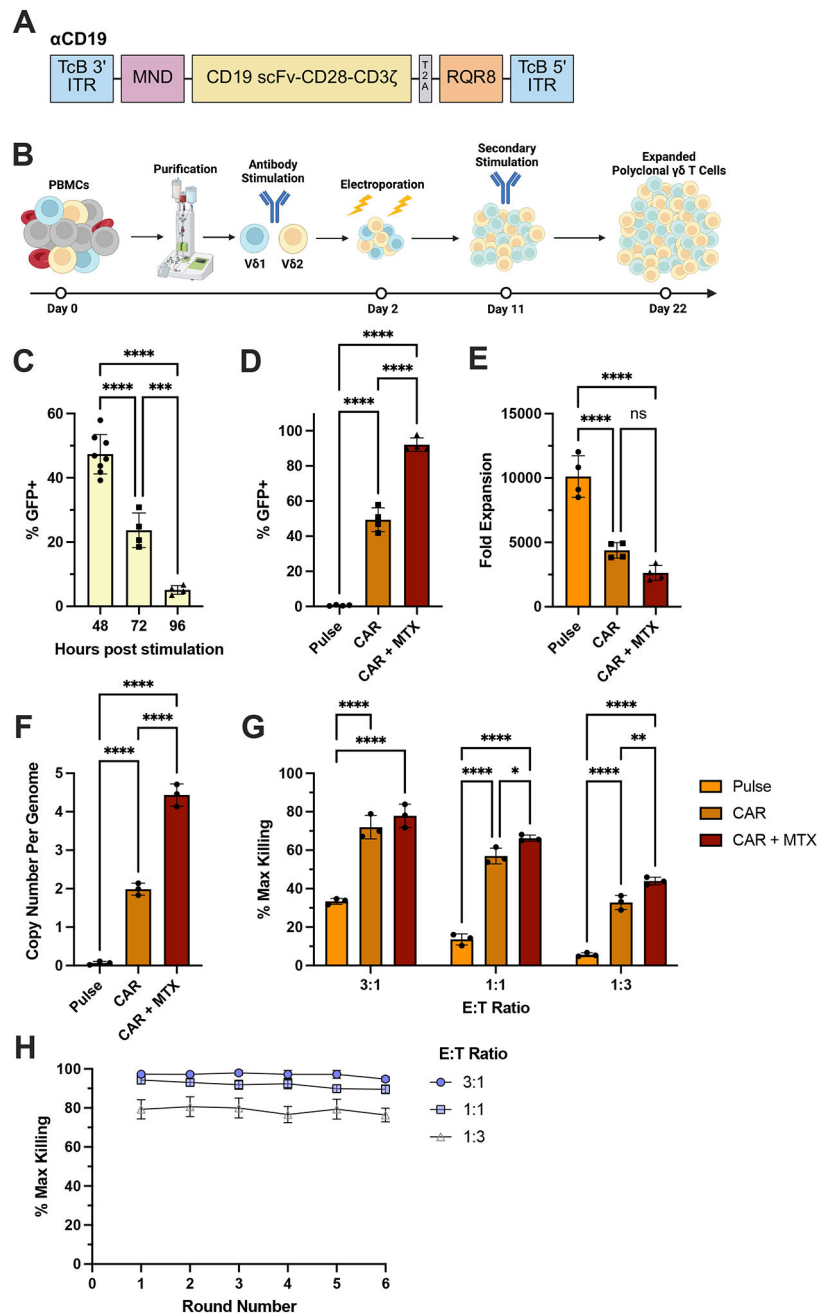


Figure 2. Non-virally engineered CAR- $\gamma\delta$ T cells display potent *in vitro* cytotoxicity.

(A) Diagram of construct containing CD19-CAR, mutant DHFR, conferring resistance to MTX in engineered cells, and GFP, used to manufacture CD19-CAR $\gamma\delta$ T cells. (B) Diagram of our clinically scalable platform for $\gamma\delta$ T cell engineering and expansion. Human $\gamma\delta$ T cells are isolated by negative immunomagnetic separation from healthy donor PBMCs, then stimulated with plate-bound α Pan- $\gamma\delta$ TCR and soluble α CD28. $\gamma\delta$ T cells are then electroporated with genome engineering reagents at day 2, re-stimulated on day 11, and collected at day 22. (C) Efficiency of non-viral construct delivery, as measured by GFP expression after stimulation and transfection of DNA-based CAR constructs alongside

hyperactive *Tc Buster* transposase mRNA (n=8 for 48-hour time point, n=4 for 72- and 96-hour time points, One-way ANOVA with Tukey's multiple comparison test). (D) Frequency of CAR- $\gamma\delta$ T cells, as measured by GFP expression, and (E) fold expansion on day 22 following MTX selection (n=4, One-way ANOVA with Tukey's multiple comparison test). (F) CAR copy number per $\gamma\delta$ T cell genome following non-viral integration as measured by ddPCR. Each point represents the average of three technical replicates (n=3, One-way ANOVA with Tukey's multiple comparison test). (G) *In vitro* cytotoxicity of Raji-luc cells after coculture with CD19-CAR $\gamma\delta$ T cells. Assays were performed at 3:1, 1:1, and 1:3 E:T ratios. Luminescence intensity was used to quantify cytotoxicity of Raji-Luc target cells, with Triton X-100 serving as a positive control for max killing (n=3 independent biological donors, individual data points represent the average of 3 technical replicates. Two-way ANOVA with Tukey's multiple comparison test). (H) Cytotoxicity of CD19-CAR $\gamma\delta$ T cells after repeated rounds of exposure to Raji-Luc target cells (n=6) (****p<0.0001, ***p<0.001, **p<0.01, and *p<0.05)

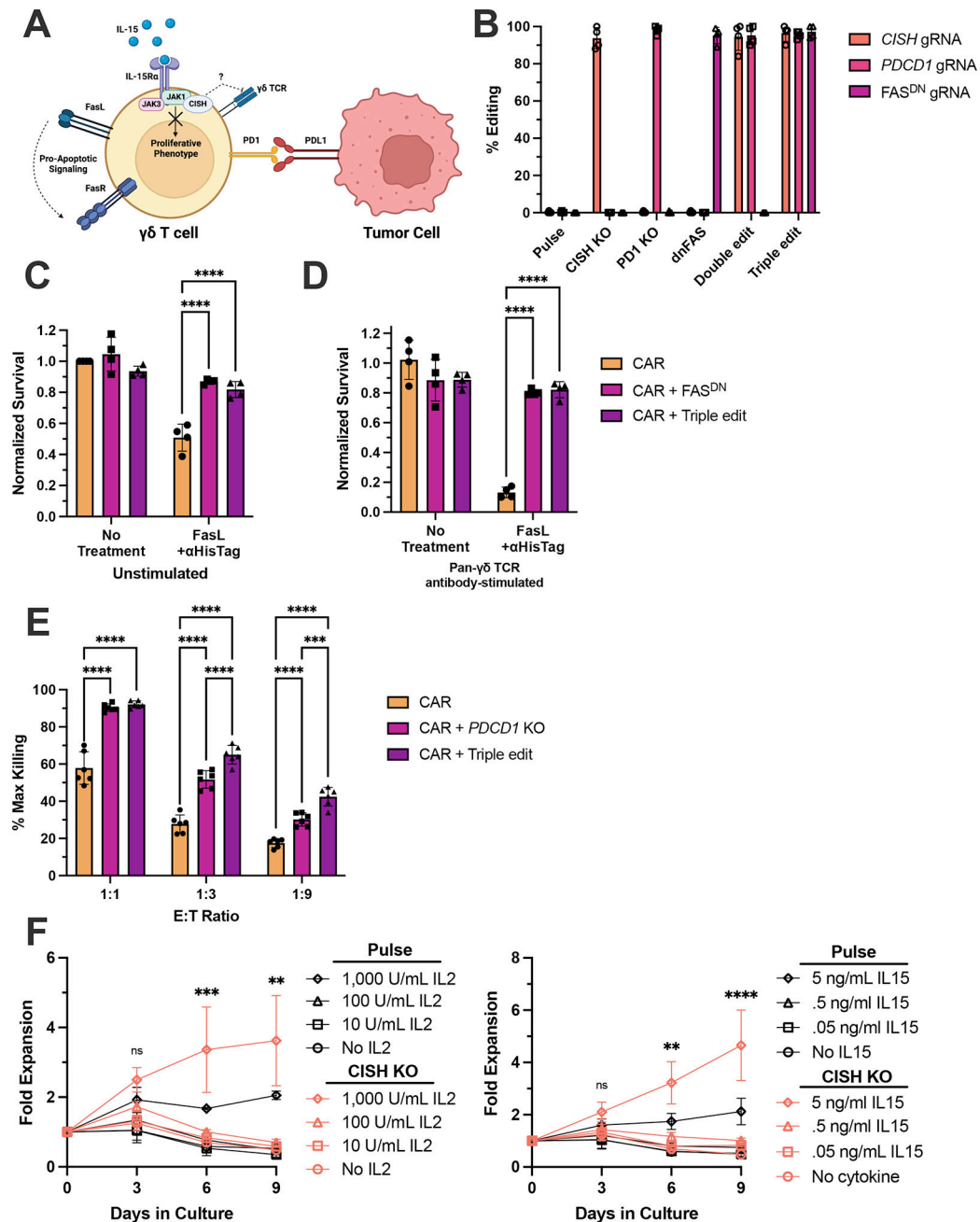


Figure 3. Multiplex gene knockout enhances $\gamma\delta$ T cell function in immunosuppressive environments.

(A) Diagram highlighting the roles of *CISH*, *PDCD1*, and *FAS* in suppressing $\gamma\delta$ T cell activity *in vivo*. (B) Editing efficiency of *CISH* KO gRNA, *PDCD1* KO gRNA, dnFAS gRNA, a combination of *CISH* and *PDCD1* gRNA, and all three gRNAs, as well as a no-gRNA control in $\gamma\delta$ T cells when cotransfected with ABE (n=4). Normalized survival of (C) unstimulated and (D) α Pan- $\gamma\delta$ TCR antibody-stimulated control, dnFAS, and triple edit CAR- $\gamma\delta$ T cells either untreated or treated with soluble Fas ligand. Live cell counts were quantified after 24 hours and normalized to untreated control CAR $\gamma\delta$ T cells (n=4, Two-way ANOVA with Tukey's multiple comparison test). (E) Cytolysis of PDL1^{high} Raji-Luc

target cells following coculture with MTX-selected CAR $\gamma\delta$ T cells, MTX-selected CAR $\gamma\delta$ T cells with a PD1 KO, and MTX-selected CAR $\gamma\delta$ T cells with PD1 KO, CISH KO, and Fas dn (n=6, Two-way ANOVA with Tukey's multiple comparison test)). (F) Fold expansion of engineered $\gamma\delta$ T cells cultured in decreasing amounts of IL2 (*left panel*) or IL15 (*right panel*) (n=4 over 2 independent biological donors, Two-way ANOVA with Tukey's multiple comparison). (****p<0.0001, ***p<0.001, and **p<0.01)

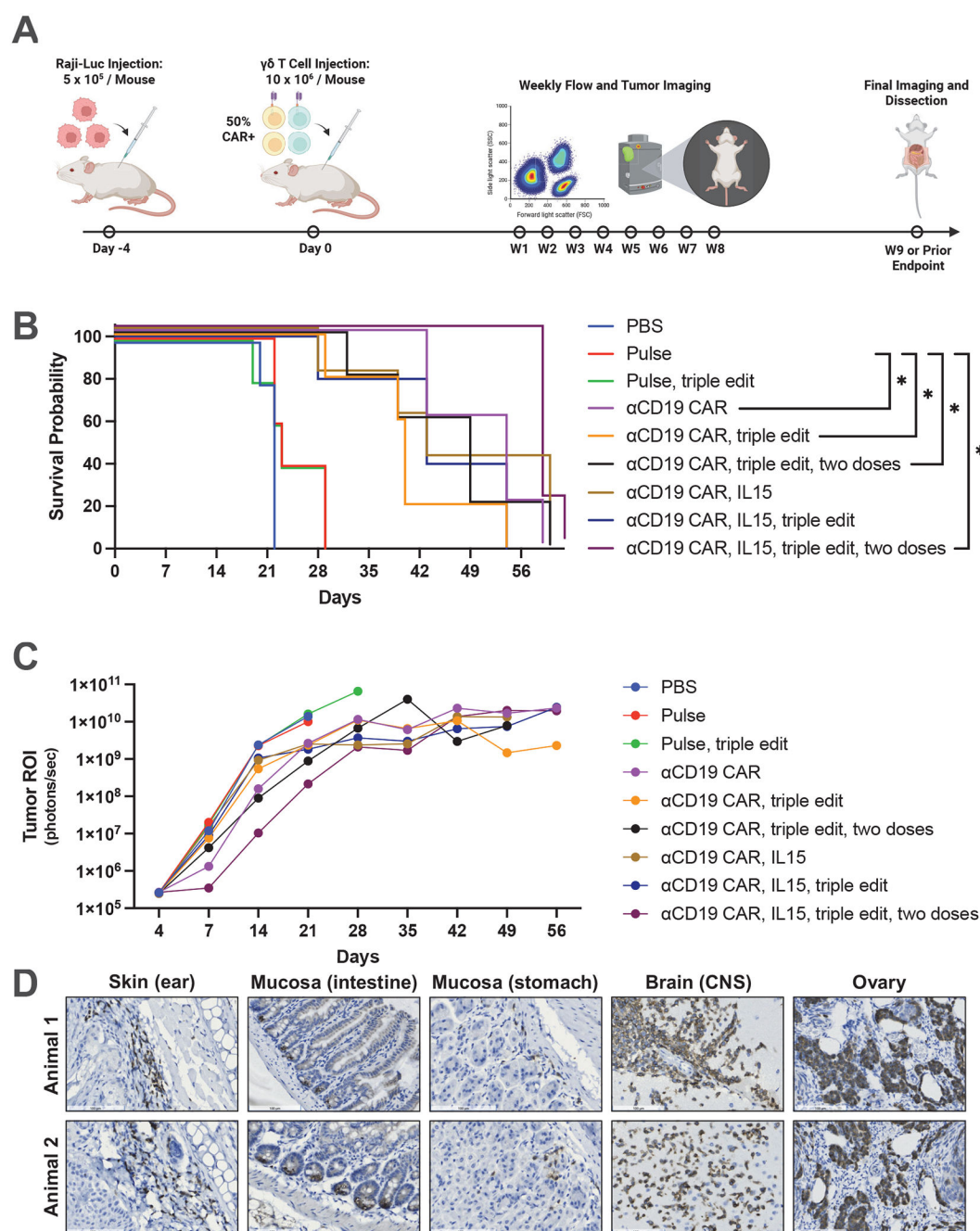


Figure 4. Engineered polyclonal $\gamma\delta$ T cells exhibit anti-tumor activity *in vivo*.

(A) Diagram of *in vivo* tumor challenge model. Mice were injected intravenously with 5×10^6 Raji-luc cells on day -4 and treated with 10×10^6 $\gamma\delta$ T cells on day 0. Mice that received two doses were injected again with 10×10^6 $\gamma\delta$ T cells on day seven. Tumor growth and overall survival was measured and flow cytometry on blood was done weekly until they reached endpoint. (B) Kaplan-Meier survival curve and (C) Tumor growth for NSG mice injected with Raji-Luc cells and treated with the indicated $\gamma\delta$ T cells ($n=5$) (* $p < 0.00714$) (see Supplemental Figure 17B and Supplemental Table 2 for statistical comparison details). (D) Immunohistochemistry staining

of CD3 on paraffin embedded skin, intestines, stomach, brain, and ovary in two animals treated with $\gamma\delta$ T cells engineered with CD19-CAR and IL15.

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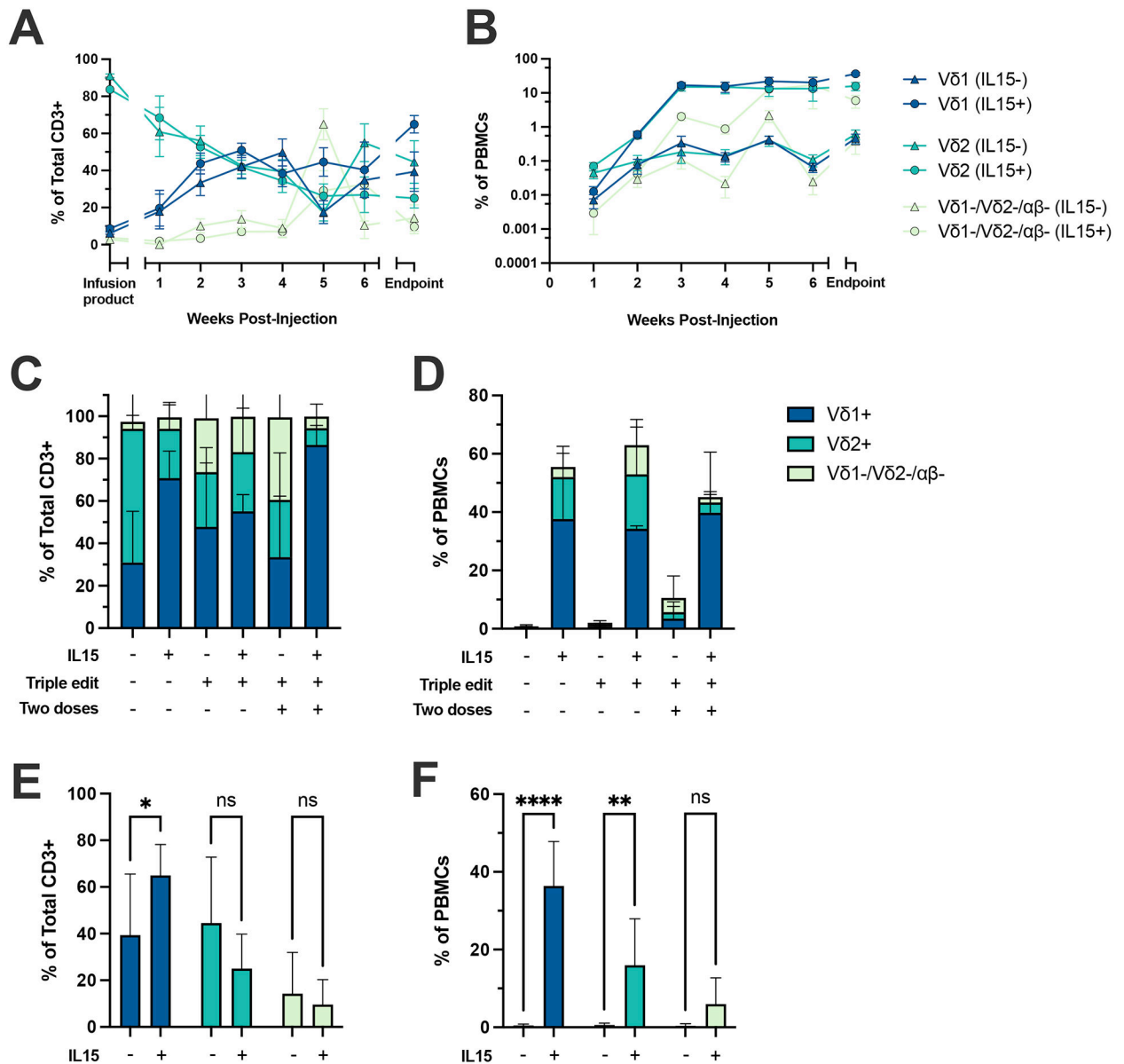


Figure 5. Effect of gene engineering on $\gamma\delta$ T cell subset frequency and persistence *in vivo*. Weekly frequency of V δ 1+, V δ 2+, and V δ 1-V δ 2- $\alpha\beta$ - subsets (A) as a percent of total CD3+ cells and (B) as a percent of total PBMC as measured by flow cytometry. “Infusion product” indicates the frequency before injection while “Endpoint” indicates the frequency at necropsy (n=10). (C) Frequency of $\gamma\delta$ T cell subsets as a percent of total CD3+ cells in PBMCs and (D) as a percent of total PBMCs at necropsy as measured by flow cytometry (n=5). (E) Frequency of $\gamma\delta$ T cell subsets as a percent of total CD3+ cells in the bone marrow and (F) as a percent of total bone marrow cells at necropsy as measured by flow cytometry (n=5). (G) Frequency of $\gamma\delta$ T cell subsets as a percent of total CD3+ cells and (H) as a percent of total PBMCs at endpoint in groups that received a single dose of cells engineered with a constructs that did not contain IL15 compared with groups that received

a single dose of cells engineered with constructs that contained IL15 (n=10, Two-way ANOVA with Tukey's multiple comparison test). (****p<0.0001, **p<0.01, and *p<0.05)

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